Preliminary Project Proposal

HEAD-TRACKING MONITOR ARM

Background and Motivation

In modern work and home environments, users spend extended periods in front of computer monitors. Static monitor stands require frequent manual readjustment to maintain a comfortable viewing angle, which can be disruptive and inconvenient. This project proposes an automated solution to enhance user comfort and create a more seamless interaction with the display. The core idea is to develop a monitor arm that actively tracks the user's head position and adjusts itself automatically, eliminating the need for manual intervention. Beyond ergonomic benefits for desktop use, this technology has significant potential for broader applications. The system could be adapted for smart televisions or portable displays, such as LG's StandbyME, allowing the screen to automatically orient itself towards the user regardless of their position in a room. This creates a more dynamic and user-centric viewing experience, representing a forward-thinking step in human-computer interaction.

Key Objectives

1. **Design & Simulation:** To complete the theoretical design of the monitor arm, including all kinematic and torque calculations, the CAD model in Fusion360, and the schematic and layout for a custom Printed Circuit Board (PCB).
2. **Software Development:** To develop a robust computer vision script in Python that uses a camera to detect a user's face and output directional control signals to a Raspberry Pi.
3. **Hardware Prototyping:** To manufacture all mechanical parts using 3D printing, fabricate and assemble the custom PCB, procure all electronic components, and construct the final scaled-down prototype.
4. **System Integration & Evaluation:** To integrate the software and hardware, calibrate the system for smooth and accurate tracking, and conduct comprehensive testing to evaluate the prototype's performance against the project aims.

Overall Aim and Objectives

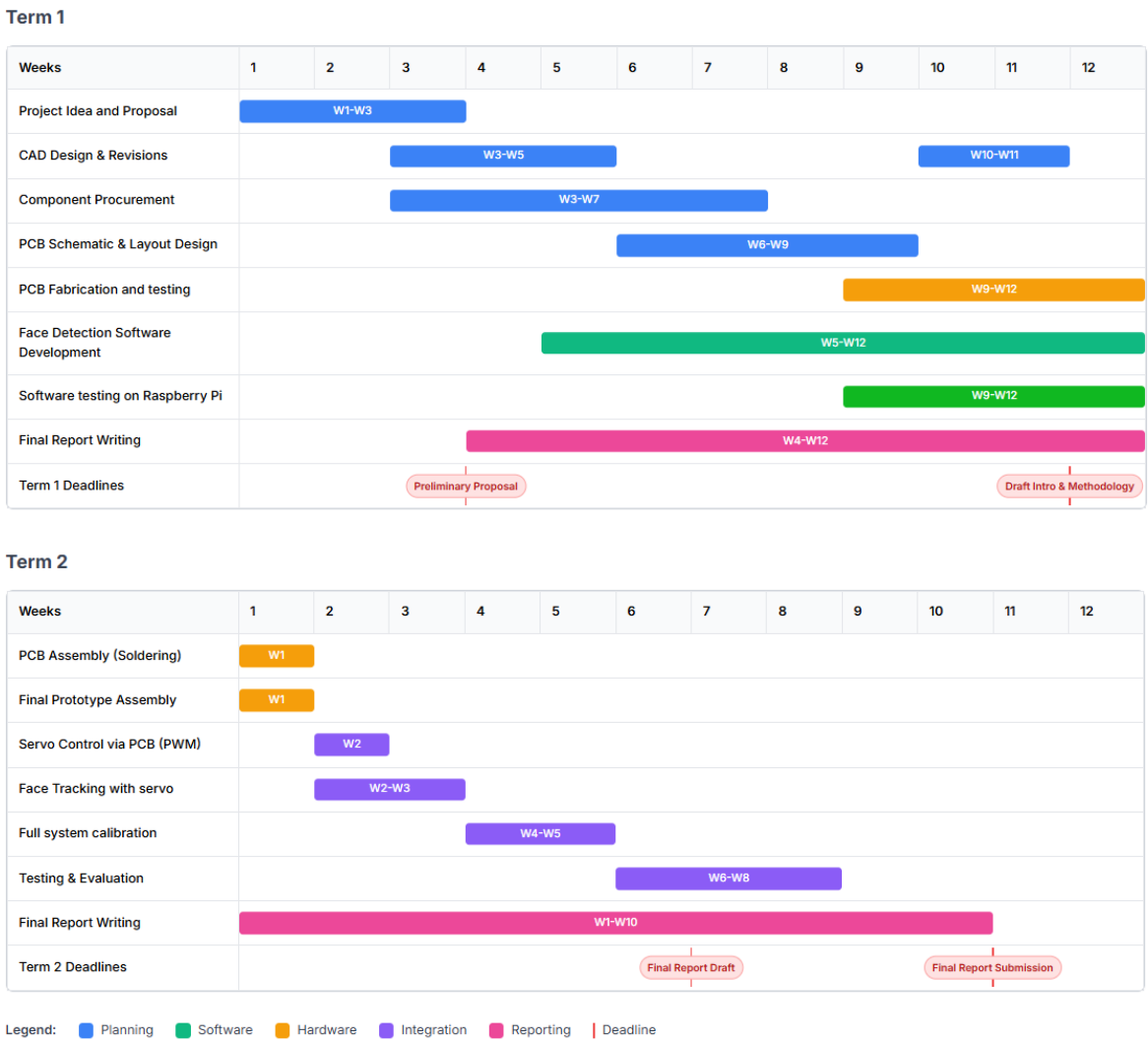
Overall Aim: To design, build, and test a scaled-down prototype of a dual-axis monitor arm that uses computer vision to track a user's head and automatically adjust its position. The prototype will be designed to hold a 12.9-inch iPad and will be constructed from 3D-printed materials to fit budgetary constraints.

Objectives:

1. **Design Phase:**
    a. Conduct calculations for forward kinematics and torque to determine the required cross-sectional area of components and select appropriate servo motors.
    b. Develop and finalize the CAD model of the monitor arm using Fusion360, including joint implementation and prismatic rail design.
    c. Conduct beam bending calculations to identify appropriate cross-sectional area of the
    d. Design the schematic and layout for a custom PCB to manage power distribution and motor control signals from the Raspberry Pi.

2. **Development Phase**:
    a. Develop the face detection software using Python with open computer vision libraries to process a camera feed and output directional signals.
    b. Program a Raspberry Pi microcontroller to interpret signals from the Python script and actuate the servo motors accordingly.
    c. Manufacture all components using 3D printing (ABS) and procure all necessary electronic components (servos, Raspberry Pi, power supply, etc.).
    d. Send PCB fabrication request then Assemble the custom PCB by soldering the electronic components onto the fabricated board. Followed by following
    e. Assemble the mechanical structure and complete the electronic wiring,
3. **Evaluation Phase:**
    a. Integrate the software and hardware components, ensuring the Python script can reliably control the monitor arm via the Raspberry Pi.
    b. Calibrate the system to ensure smooth and accurate tracking of the user's head.
    c. Test the fully assembled prototype against the predefined criteria, documenting its performance and identifying areas for improvement.

Gantt Chart



Project Title:	Head-tracking monitor arm						Submission Date:	16th October 2025	
Student Name:	Junbyung			Park					
Project Risk	Severity			Potential			Score (Severity x Potential)	Mitigation Measures	
	L	M	H	L	M	H	L=1, M=2, H=3		
Structural Failure of 3D-Printed Parts			H		M		6	-Print components with a high infill percentage (>50%) for increased strength. - As planned, build and test a small-scale model to validate the design's integrity.	
Inaccurate or Slow Face Tracking Performance		M				H	6	- Have a fallback plan to test mechanics, such as tracking a simple, high-contrast object (e.g., an ArUco marker). -Use pre-made AI Camera and their libraries to track the User.	
Insufficient Servo Motor Torque		M			M		4	- Perform detailed torque calculations based on the final CAD model's mass and lever arms. - Select servos with a torque rating at least 1.5x the calculated requirement.	
Delayed Delivery of Electronic Components	L					H	3	- Identify all components (Raspberry Pi, servos, camera, power supply) and order them as early as possible. - The project Gantt chart includes a buffer period between the scheduled end of procurement and the start of assembly to absorb potential shipping delays.	
Power Supply Unit Cannot Provide Enough Current		M		L			2	Calculate the maximum current draw of all servos moving simultaneously. - Select a power supply unit (PSU) that can provide at least 1.5x the calculated peak current.	
Algorithm Confused by Environment			H			H	9	Implement logic to handle cases where multiple faces are detected (e.g., track the largest or most central face).	
3D Printed Parts Don't Fit		M			M		4	- buffer period between CAD Design and Hardware Assembly - Design parts in CAD with appropriate tolerances for assembly	
Budget overrun	L			L			1	- Include a 10-15% contingency fund in the budget for unexpected costs. - Utilize university resources (e.g., 3D printers, lab equipment) to minimize expenses.	